EventOps Intelligence

Production-assessed structured RAG and bounded cross-source verification | v.01 | October 2026

Runtime architecture

Knowledge / RAG

CSV + JSON > FastEmbed > pgvector > Query filters > Cited answer

Verification

Calendar/Email/Web > LangGraph > Rules + refs > PostgreSQL > CSV / replay

React/Vite + React Router + React Flow → FastAPI → PostgreSQL/pgvector. Both workflows share persisted reference data, but neither treats generated text as verified identity or eligibility.

Authoritative dataset and retrieval contracts

Dataset / owner	Indexed records	Data authority
events.csv	3	Event specifications, identifiers and dates
policies.csv	9	Check-in, registration and operational policy
directory.json	7	Participant identity, registration and status
Persistent vector index	19 x 384	Local FastEmbed vectors in PostgreSQL / pgvector

Knowledge path. Typed ingestion → local embedding → filtered vector retrieval → context resolution → optional Gemini JSON generation → citation-ID validation. Query planning preserves named EVT IDs, policy categories and explicit user wording. Ambiguous follow-up references fail.
Verification path. Bounded public HTTPS tool reads → extraction → normalization → deduplication → authoritative reference checks → discrepancy flags → persisted run/results. LangGraph provides deterministic stage sequencing, not open-ended LLM tool autonomy.

Core API surface

Method / path	Contract / behaviour
POST /api/rag/retrieve	Question + top_k; returns query plan, ranked evidence and provenance
POST /api/conversations/{id}/messages	Persist conversation, retrieved sources, optional generated answer and status
GET /api/knowledge/sources	Indexed documents, counts and source provenance
POST /api/agents/runs	202; event_id + max_records; bounded background verification trigger
GET /api/agents/runs/{id}[/results]	Persisted stage execution, summaries and field-level results
GET /api/agents/runs/{id}/export.csv	All or verified-only records; CSV does not mutate the completed run

Evidence, controls and limitations

Implementation evidence from the repository acceptance reports, 11 October 2026.

Production assessment: observed results

System	Verified technical observation
Frontend / backend	Render static site + Python API; individually deployed same implementation; health/readiness reported passing
Persistence	Neon PostgreSQL with pgvector; 4 migrations, 19 reference rows and 384-dimensional vectors retained
Cross-source run	13 parsed/normalised observations; 7 saved results; 1 verified, 6 flagged; 6 duplicates merged
Positive / conflicts	Controlled Mailtrap observations verified active, duplicate, inactive and name-conflict cases with field evidence
Grounded generation	One bounded real Gemini answer with matching retrieved policy citation; returned to disabled afterward
Exports / replay	All and verified-only CSV; saved run history/replay and restored generation-status boundaries

Safety and data integrity invariants

Boundary	Enforcement / current position
Source authority	CSV/JSON reference tables determine eligibility; calendar/email/web supply untrusted observations
Provenance	Original source URL/IDs, normalized fields, per-field discrepancies, run timestamps and status preserved
Read-only tools	Public HTTPS; time/size limits; no private/local URLs, redirects, email sending or arbitrary tool execution
Reviewer isolation	Demo-token scoped conversations/runs; exact-origin CORS; rate and request budgets; not enterprise identity
Generation	No model call if disabled/unsupported; up to 5 citations; unsupported IDs rejected; source evidence survives failure
External automation	n8n service handoff endpoints/workflow prepared but not configured or proven live

Recorded verification

Validation track	Result / interpretation
Backend + frontend	217 Python tests passed, 1 live opt-in skip; 16 frontend unit tests passed; frontend build passed
Browser (release)	25 Playwright passes and 3 opt-in skips in the recorded release assessment
Post-submission checks	29 Playwright passes and 3 opt-in skips recorded in the original checkout after Phase 1
Production checks	Real Mailtrap, public Calendar/Web observations, persisted CSV/history, cited RAG and bounded Gemini demo

Not certified as production-grade: demonstration-token sessions are not full authentication; FastAPI BackgroundTasks are not a durable restart-resuming queue; free-host cold starts and memory headroom are not independently verified; optional n8n automation remains unconfigured. The event/participant dataset is synthetic.

Technical sources: README.md; docs/deployment/DEPLOYMENT_STATUS.md; docs/deployment/PRODUCTION_ACCEPTANCE_REPORT.md; docs/iterations/post-submission/TEST_EVIDENCE.md; docs/decisions/ADR-001-source-authority.md; backend/app/{agents,rag,verification,api}.